SWAMY R

Full Stack Developer | React.js • Python • FastAPI • PostgreSQL

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PROFESSIONAL SUMMARY

Full Stack Developer with 2+ years of hands-on experience building production-grade React.js applications and FastAPI/PostgreSQL backends for real-time industrial monitoring systems, enterprise dashboards, and RFID-based tracking software. Skilled in configuration-driven UI architecture, REST API design, WebSockets, TypeScript, role-based access control, and end-to-end deployment using Docker and CI/CD. Proven ability to independently own modules from requirement analysis through production deployment, and to collaborate closely with engineers and scientists to deliver reliable, scalable software.

CORE SKILLS

Languages: Python, JavaScript, TypeScript

Frontend: React.js, Vue.js, Vite, Ant Design, Tailwind CSS, HTML5, CSS3

Backend & APIs: FastAPI, Flask, Node.js, REST APIs, WebSockets, JWT Authentication, SQLAlchemy, Alembic, Pytest

Databases & Storage: PostgreSQL, MongoDB, MinIO

AI & LLM Integration: Ollama, Qwen 2.5, Llama 3, RAG (Retrieval-Augmented Generation), Local/Private LLM Deployment, Prompt Engineering

Visualization & Reporting: Recharts, Plotly, Chart.js, jsPDF

Cloud & Integrations: Supabase, Cloudinary, OneSignal, RFID Systems

DevOps & Tools: Git, Docker, CI/CD, Linux

Core Concepts: Configuration Management, Real-time Data Handling, Role-Based Access Control, API Integration, Predictive Maintenance

PROFESSIONAL EXPERIENCE

Project Associate (Software Developer), CMTI (Central Manufacturing Technology Institute) **Sep 2024 – Present**

Bengaluru, India

- Developed the frontend and backend of a Condition Monitoring & Vibration Analysis application (React.js, Python/FastAPI) integrated with OROS DAQ and NV Gate hardware, supporting acceleration, velocity, displacement, FFT and RMS analysis for fault detection (misalignment, cage defects, inner/outer race bearing defects).
- Built and enhanced the NIAMT Smart Manufacturing Dashboard (React.js, FastAPI, PostgreSQL) to monitor temperature, energy, pressure and level parameters in real time, with automated email/message alerts on threshold breaches for NIAMT machines.
- Designed and independently deployed a Machine-Hour Maintenance & Job Planning system for SMMDC covering 7 machines, with role-based workflows across 3 roles (Admin, Supervisor, Operator), machine/job-hour cost reporting, tracking of active jobs and machine usage, and PDF/CSV export.
- Developed a Manpower Scheduling & Mapping application for service allocation, availability checks and cost generation across Advanced Material Characterization teams, allowing manpower users to view scheduled jobs and add remarks after service completion.
- Contributed core modules to an enterprise Project Planning & Management (PPM) platform (React, Vite, Ant Design, Tailwind CSS, FastAPI, SQLAlchemy, PostgreSQL, Alembic, MinIO) covering proposals, milestones, payments, documents, communication and analytics, with JWT-secured role-based access for 6 user roles (Admin, Group Head, Centre Head, Scientist, Director, Guest) serving 40+ users, including 30 scientists, 10 admins and 1 director; containerized and deployed with Docker for active production use.
- Designed and deployed a local, private LLM pipeline using Ollama (Qwen 2.5 / Llama 3) integrated into FastAPI for enterprise proposal intelligence and automated documentation within the PPM platform.
- Engineered a RAG-style Context Engine (ContextBuilder) that aggregates live relational database context (financials, remarks, customer data, and milestones) to inject structured ground truth into LLM prompts.

- Building the live and historical vibration-monitoring dashboard for the ongoing CMF Vibration Monitoring System as part of a team, enabling machine defect analysis for engineering teams.
- Provide ongoing application maintenance and support for the SMMDC E-Forecast Maintenance system, and support welding-defect analysis through image annotation of defects such as porosity and displacement.
- Configured CI/CD pipelines in Git and containerized services with Docker, automating testing and deployment across multiple applications; partnered with senior engineers and scientists to debug, optimize and stabilize configuration-driven workflows, improving overall system reliability and performance.

PROJECTS

Vibration Monitoring System | *Python, React.js, FastAPI, PostgreSQL, WebSockets, Recharts*

- Built application modules for real-time vibration data processing and visualization; validated and optimized backend logic to ensure stable, accurate data handling.

Smart Workshop Dashboard | *MERN Stack, WebSockets, JWT, Plotly*

- Designed and built role-based, configurable dashboard features using full-stack technologies; integrated backend services and APIs to support automation and real-time updates.

Maintenance Tracking System | *PERN Stack, jsPDF, Chart.js*

- Delivered end-to-end application workflows from requirement analysis to deployment; improved reporting accuracy by strengthening data validation and backend processing.

RFID-Based File Tracking System | *TypeScript, FastAPI*

- Developed software to track physical files using RFID, identifying the real-time location of each file; built the backend with FastAPI and a type-safe frontend with TypeScript for reliable location tracking.

KLS Mart — Gold Collection & Gold Chitti Platform (Independent Project) | *React.js, FastAPI, Supabase, Cloudinary, OneSignal*

- Developed a full-stack website for KLS Mart to manage gold collection and gold chitti (chit fund) operations end to end.
- Used Supabase and Cloudinary for scalable data and image storage, and OneSignal for real-time push notifications; built a responsive frontend and backend APIs to support complete business workflows.

EDUCATION

Bachelor of Engineering, Computer Science and Engineering

RajaRajeswari College of Engineering, Bengaluru, India | Dec 2020 – Jul 2024 | CGPA: 7.61

ADDITIONAL INFORMATION

- Comfortable owning a module end-to-end — from requirement analysis and API design through frontend implementation, testing, containerization and production deployment.
- Experienced working directly with senior engineers and scientists to translate real-world industrial requirements (sensor data, machine parameters, RFID hardware) into reliable software.